

Scenario S3-O2-I2 — Our Level 2 (Ideal) / Investor Level 2 (Ideal)

PRIMARY RECOMMENDED PATH

Scenario Summary

The balanced early-capital path. The founders self-fund at ideal intensity (~100% of baseline spend) through Pre-Pilot and Pilot 1, producing a credible live pilot with real patients, measured throughput, and operational data over ~6 months. An ideal-level investor arrives right after Pilot 1 (approximately M6-M7), providing properly-sized capital to fund Pilot 2 and Production Readiness at sustainable resource levels. Both parties invest at the baseline needed to meet milestones without excess. This combination produces the strongest investor narrative ("we built it, tested it with real patients, proved demand, and are raising to scale — here is our Pilot 1 data") while keeping founder capital exposure in a manageable range.

This is the recommended path for Health Hub because it optimizes across all six planning dimensions simultaneously: timeline, cost, founder exposure, dilution, risk, and narrative quality.

- **Economic cost to Production:** \$170,000-215,000 / INR 14,110,000-17,845,000
- **Target timeline to Production:** 18-22 months
- **Founder/company capital before investor:** \$53,000-65,000 / INR 4,399,000-5,395,000
- **Investor capital modeled:** \$117,000-150,000 / INR 9,711,000-12,450,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Break-even month:** 22-28
- **Risk level:** Medium
- **Recommendation:** Pursue (Primary)
- **Main risk to watch:** investor close timing — must happen within 45 days of Pilot 1 completion to maintain momentum and avoid a capital gap

Why This Path

The Core Logic

S3-O2-I2 is the intersection of four constraints:

1. **Minimum credible founder investment.** \$53,000-65,000 over 6 months is enough to demonstrate genuine commitment to investors without bet-the-house exposure. At ~\$10,000-13,000/month, the founder is funding a real team at real utilization — not a side project.
2. **Maximum investor data before capital deployment.** The investor enters after Pilot 1, meaning they see real patients, real consultations, real partner engagement, and real operational data before writing a check. This is the strongest possible proof point short of revenue traction.
3. **Right-sized scaling capital.** \$117,000-150,000 funds 12-16 months of Pilot 2 + Production Readiness at ideal (not bloated) levels. The operational staffing ladder, infrastructure, and development team all run at the capacity needed to meet milestones — no more, no less.
4. **Timeline that respects both urgency and quality.** 18-22 months is fast enough to maintain team energy and competitive positioning, but slow enough to build the operational backbone (SOPs, training, compliance, partner management) that a healthcare platform requires.

Why Not Other Paths

Alternative	Why S3-O2-I2 Is Better
S1-L2 (self-fund everything)	Costs the founder \$155,000-195,000 vs. \$53,000-65,000. Same risk profile but 3x the personal financial exposure. No investor network or validation.
S2-O2-I2 (investor later)	\$18,000-30,000 more founder capital for 2-4 more months. The delay does not produce proportionally more investor leverage.
S3-O2-I1 (bootstrapped investor)	Same founder cost but investor under-funds scaling. Cash goes negative by M18. High risk of operational failure during the most critical growth phase.
S3-O2-I3 (fully funded investor)	\$50,000-70,000 more total spend for 2-4 months of acceleration. Potentially exceeds the 20% investor pool. Good as an upside if Pilot 1 exceeds expectations, but not the base plan.
S3-O1-I2 (bootstrapped founder)	Under-invests in Pre-Pilot and Pilot 1, producing a weaker pilot result. The investor sees less traction and demands more equity or passes entirely.

Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M4	4 months	Architecture freeze (M1), APK backlog lock (M2), doctor training draft (M3), admin callback SOP (M3), internal rehearsal cycles (M3-M4), investor pitch deck draft (M4).	Core web flows stable, internal rehearsals pass, doctor and admin runbooks versioned, APK core ready, investor materials prepared.
Pilot 1	M4-M6	2 months	Warm-network onboarding (M4-M5), first paid consults (M5), callback calibration, doctor playbook v1 live, investor conversations active, Pilot 1 data collection.	Ethiopia pilot runs live for 2 months with measured throughput; >200 active users; >50 paid consults; investor term sheet signed or in final negotiation.
Pilot 2	M6-M12	6 months	Investor capital deployed (M7), marketing push (M8), Kenya prep (M9), partner reporting v2 (M10), roster scaling (M11), mid-pilot review (M9).	5k-10k user-capable operations with full APK, clean reporting, staffed callback coverage, Kenya market research complete.
Production Readiness	M12-M20	8 months	Compliance closeout (M14), security hardening (M15), release freeze (M16), launch playbooks (M17), certification refreshers (M18), production support schedule (M19), final QA (M20).	Current prototype scope is production-safe across web and patient mobile with documented fallback operations.

Milestone Criteria (Detail)

Each phase gate has explicit pass/fail criteria. These are not aspirational — they are hard gates.

Pre-Pilot Gate (M4):

- Web platform: all 10 feature modules load without error; auth flow complete; SSR stable
- APK: core patient journeys functional (registration, dashboard, consult request)
- Ops: doctor training playbook v1 complete; admin callback SOP v1 complete; 2+ internal rehearsal cycles passed
- Business: investor pitch deck drafted; 3+ partner clinics signed for Pilot 1

Pilot 1 Gate (M6):

- Users: >200 active patients registered; >50 completed paid consultations
- Operations: callback response time <4 hours; no unhandled patient escalations; doctor utilization >40% during operational hours
- Technical: zero P0 bugs in production; uptime >97%; SSR stable under real load
- Revenue: >\$400/month from real transactions (subsidized pricing acceptable)
- Investor: term sheet signed or in final negotiation with <30 days to close

Pilot 2 Gate (M12):

- Users: >2,000 active patients; >500 monthly consultations
- Operations: full staffing ladder active; callback SOP v2 in use; partner reporting automated
- Technical: APK feature-complete; web hardening 90%+ complete; load-tested for 5,000 concurrent users
- Revenue: >\$3,000/month with clear growth trajectory
- Kenya: market research complete; 2+ potential partners identified

Production Readiness Gate (M20):

- Compliance: all regulatory documentation filed; partner-license model validated
- Security: penetration testing complete; OWASP Top 10 addressed; security incident response plan documented
- Operations: near-24/7 coverage achieved (3-shift model); all SOPs at v3+; support ticketing system live
- Release: release governance process documented and tested; rollback procedures validated
- Revenue: >\$8,000/month; monthly revenue growth >10%

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at ideal pace; APK build running in parallel with web; doctor/admin training materials drafted and rehearsed; support desk scripts and callback trees defined; partner onboarding kits for Pilot 1 clinics; investor pitch deck with financial model.
Pilot 1	Patient app covers core journeys (registration, dashboard, GP consult, basic specialist referral); provider web stable for GP and specialist intake; doctor roster active; admin callbacks operational; manual fulfilment tracked in-system; Pilot 1 data dashboard for investor conversations.
Pilot 2	All patient features complete (prescriptions, labs, diagnostics, notifications, payments); partner dashboards functional; staffing ladder at ideal levels; reporting and payment loops operational with increasing automation; Kenya market research and partner outreach; APK hardened and submitted to Play Store.
Production Readiness	Current prototype scope production-grade across web and patient mobile; trained ops staff at near-24/7 coverage; compliance documentation complete; release governance in place; support escalation paths tested; incident response procedures documented.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	95.0%	Next business day	3 core team at ideal utilization + 7 part-time developers at ~480 hrs/mo + 2 advisors; no employed clinical coverage, only training and rehearsals.
Pilot 1	97.0%	Same-day for critical issues	Minimum employed coverage: 2 doctors, 2 admin ops, 2 technical support. Partner clinics supplement.
Pilot 2	98.0%	4-hour P1 response	Full employed coverage: 4 doctors, 4 admin ops, 2 technical support. Callback and partner reporting loops active. Ideal staffing enables proper coverage.
Production	99.0%	2-hour P1 response	Full employed coverage: 6 doctors, 6 admin ops, 3 technical support for near-24/7 coverage. Full support ticketing and escalation chain.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	\$81,700 / INR 6,781,100
Operating expenditure before Production launch	\$110,800 / INR 9,196,400
Total economic spend to Production	\$192,500 / INR 15,977,500

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes
Product and founder office	\$42,240	INR 3,505,920	3 core team at \$20/hr x 88 hrs/mo. M0-M6 at ideal utilization (\$5,280/mo x 6 = \$31,680). M7-M20 at moderate utilization as ops workstreams take priority (\$760/mo x 14 = \$10,560).
Web platform hardening	\$14,900	INR 1,236,700	745 hours at \$20/hr senior rate. Bulk completed in Pre-Pilot and Pilot 1 at ideal pace. Remaining hardening in Pilot 2.
Patient APK build	\$7,200	INR 597,600	Fixed Android quote: 480 hours at \$15/hr. Runs parallel to web in Pre-Pilot. Core ready by Pilot 1; full scope by mid-Pilot 2.
QA, DevOps, security	\$8,800	INR 730,400	~440 hours at \$20/hr. Full ideal-level QA. Includes penetration testing, load testing, CI/CD hardening.
Admin and care operations	\$87,640	INR 7,274,120	Employed doctors, admin callback staff, and technical support ladder across 16 months of live operations at ideal staffing levels. See ops staffing breakdown below.
Partner onboarding and provider success	\$9,750	INR 809,250	650 hours at \$15/hr blended. Doctor training, admin SOPs, partner reporting, onboarding kits, stakeholder management.
Legal, accounting, compliance	\$5,200	INR 431,600	Friendly-network pricing at economic value. Includes regulatory filing support for Ethiopia and Kenya.
Infrastructure, phones, and tools	\$19,770	INR 1,640,910	Hosting, devices, telecom/call costs, operational tooling across 20 months. Scales properly with phases. See infra breakdown.
Marketing, travel, rehearsals	\$9,500	INR 788,500	Pre-Pilot: internal. Pilot 1: warm-network. Pilot 2: targeted marketing push. Production: sustained acquisition. Includes 2-3 Ethiopia travel cycles.
Contingency and buffer	\$30,000	INR 2,490,000	~16% buffer on total. Adequate for medium-risk path. Covers one major unexpected cost event.

Operational Staffing Cost by Phase

Phase	Duration	Doctors	Admin Ops	Tech Support	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M4)	4 months	0	0	0	\$0	INR 0	\$0
Pilot 1 (M4-M6)	2 months	2	2	2	\$2,482	INR 206,006	\$4,964
Pilot 2 (M6-M12)	6 months	4	4	2	\$4,361	INR 361,963	\$26,166
Production Readiness (M12-M20)	8 months	6	6	3	\$6,542	INR 542,986	\$52,336
Total operational staffing							\$83,466

Operational Staffing Detail

Role	Monthly Salary (INR)	Monthly Salary (USD)	Pilot 1 (2)	Pilot 2 (4/4/2)	Production (6/6/3)
MBBS Doctor (telemedicine)	60,000	\$720	2 = \$1,440	4 = \$2,880	6 = \$4,320
Admin Ops / Callback Staff	18,000	\$217	2 = \$434	4 = \$868	6 = \$1,302
Technical Support (L1/L2)	25,000	\$301	2 = \$602	2 = \$602	3 = \$903
Phase monthly total			\$2,476	\$4,350	\$6,525

Infrastructure Cost by Phase

Phase	Duration	Tech Infra (USD/mo)	Ops Infra (USD/mo)	Combined (USD/mo)	Phase Total (USD)
Pre-Pilot	4 months	\$100	\$10	\$110	\$440
Pilot 1	2 months	\$230	\$165	\$395	\$790
Pilot 2	6 months	\$450	\$240	\$690	\$4,140
Production Readiness	8 months	\$850	\$300	\$1,150	\$9,200
One-time equipment (phones, laptops, misc)	—	—	—	—	\$1,200
Total infrastructure					\$15,770

Infrastructure Detail

Component	Pre-Pilot	Pilot 1	Pilot 2	Production
Hosting (Render)	\$50	\$115	\$225	\$400
Video (Daily.co)	\$10	\$50	\$125	\$250
Database (PostgreSQL)	\$20	\$40	\$75	\$150
Monitoring/logging	\$20	\$25	\$25	\$50
Phone top-ups/SIM	\$0	\$45	\$50	\$60
Telecom/VoIP	\$0	\$75	\$100	\$120

Call center software	\$0	\$0	\$50	\$70
Support ticketing	\$0	\$0	\$15	\$30
Misc ops tools	\$10	\$45	\$25	\$20
Monthly total	\$110	\$395	\$690	\$1,150

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~100% effective utilization throughout	Single owner/founder + 2 founding members. Parallel task execution throughout. Post-investor, founder time shifts from fundraising to operations oversight.
Technical delivery (M0-M6)	7 part-time developers, modeled at ~480 productive hrs/mo	Ideal capacity. Web and APK run in parallel. Architecture freeze by M1 enables focused execution.
Technical delivery (M7-M20)	7 part-time developers, modeled at ~480 productive hrs/mo	Investor capital maintains ideal capacity through scaling phases. No velocity cliff.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review. Increased engagement during investor negotiation (M5-M7).
Employed care coverage	Pilot 1: 2 doctors / Pilot 2: 4 / Production: 6	Full ideal ladder per params.md. Partner clinics supplement but are not relied upon for core coverage.
Employed admin ops	Pilot 1: 2 admins / Pilot 2: 4 / Production: 6	Full ideal ladder. Callbacks, exception handling, manual fulfilment.
Technical support	Pilot 1: 2 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M6)	~\$14,830	~INR 1,230,890	100% of ideal rate card. Full parallel capacity.
Post-investor (M7-M20)	~\$14,830	~INR 1,230,890	Investor capital maintains ideal rate card. No drop in velocity. This is the critical advantage of I2 over I1.

Why Sustained Velocity Matters

In S3-O2-I1, dev team capacity drops from 480 to 200 hrs/mo when the investor enters — a 58% velocity cliff at exactly the moment scaling demands are highest. In S3-O2-I2, the investor funds the team at the same ideal rate the founders established, meaning:

- Feature development continues at Pre-Pilot pace throughout Pilot 2
- APK hardening and web security work happen in parallel, not sequenced
- Bug fixes and operational improvements do not compete with new features
- The team maintains rhythm and momentum instead of context-switching to survival mode

This sustained velocity is the single largest difference between I1 and I2, and it is worth the additional \$33,500-55,000 in investor capital.

Revenue and Pricing

Revenue Sources by Phase

Revenue Source	Pilot 1 (M4-M6)	Pilot 2 (M6-M12)	Production (M12-M20)
GP consults	Subsidised, Ethiopia-first, ~\$2-3 per paid interaction	Full Ethiopia pricing at \$3.50; Kenya pricing tests at \$5	Core revenue line at full pricing
Specialist consults	Minimal (referral flow testing)	Growing after referral flow stabilizes; \$9-12 per consult	Significant margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable; 10% take-rate on growing order volume	Scales with partner conversion and user growth
Diagnostics commission	Small, admin-assisted	Improving with order flow consistency; 12% take-rate	Retention lever and cross-sell driver
Travel/care coordination	Rare, high-touch	Low-volume, higher-value (\$10-15/case)	Strategic differentiator for diaspora and medical tourism

Pricing Benchmarks

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	\$3.50 / ETB 200	\$5.00 / KES 650	Subsidised at ~60% in Pilot 1; full price from mid-Pilot 2.
Specialist consult	\$9.00 / ETB 513	\$12.00 / KES 1,560	Pilot 2 onward. Higher margin but lower volume.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward. Average order ~\$8-15 in Ethiopia.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward. Average order ~\$10-25 in Ethiopia.
Travel/care coordination	\$10-15 per case	\$10-15 per case	Small volume, high-touch. 5-15 cases/month at scale.

Subscription	Deferred	Deferred	Not modeled before meaningful scale.
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Revenue Ramp Model

The revenue model below is conservative. It assumes linear ramp within each phase and no viral or referral-driven hockey stick. Revenue from Kenya is modeled at zero during Pilot 2 and enters only in Production.

Month	Active Users (est.)	Paid Consults/mo	Pharmacy Orders/mo	Diagnostics Orders/mo	Monthly Revenue (USD)
M5	100	20	10	5	\$150
M6	250	60	25	12	\$450
M7	400	100	40	20	\$850
M8	600	160	60	30	\$1,350
M9	900	250	90	45	\$2,100
M10	1,200	350	120	60	\$2,900
M11	1,500	450	150	75	\$3,600
M12	1,800	550	180	90	\$4,200
M13	2,200	700	220	110	\$5,200
M14	2,600	850	260	130	\$6,200
M15	3,000	1,000	300	150	\$7,200
M16	3,500	1,200	350	175	\$8,500
M17	4,000	1,400	400	200	\$9,800
M18	4,500	1,600	450	225	\$11,000
M19	5,000	1,800	500	250	\$12,500
M20	5,500	2,000	550	275	\$14,000

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	\$8	\$12	\$18
Gross margin	40%	50%	60%
Monthly ARPU	\$2.0	\$3.2	\$5.0
Monthly churn	12%	9%	6%
LTV/CAC target	1.6x	2.8x	3.5x
Doctor utilization	40%	55%	65%

Revenue Composition at Production (M20)

Stream	Monthly Revenue	% of Total
GP consults	\$7,000	50%
Specialist consults	\$3,600	26%
Pharmacy commission	\$1,650	12%
Diagnostics commission	\$1,100	8%
Travel/care coordination	\$650	4%
Total	\$14,000	100%

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M4)	\$5,300	INR 439,900	1.5 months at Pilot 1 burn	Investor pipeline must be warm. If no warm leads by M3, begin fallback planning.
Pilot 1 (M6)	\$4,680	INR 388,440	0.8 months	Critical: investor close must happen within 30 days. Term sheet must be signed by M6.
Post-Close (M7)	\$122,430	INR 10,161,690	10+ months	Healthy runway. Execute Pilot 2 plan.

Pilot 2 (M12)	\$74,890	INR 6,215,870	5.5 months	Comfortable. On track for Production Readiness.
Production Readiness (M20)	\$22,390	INR 1,858,370	1.5 months + revenue exceeding burn	Revenue should be at or near covering monthly costs. Remaining cash is contingency buffer.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	\$13,000	\$0	\$0	\$11,650	\$1,350	\$1,350
M2	Pre-Pilot	\$13,000	\$0	\$0	\$11,650	\$1,350	\$2,700
M3	Pre-Pilot	\$13,000	\$0	\$0	\$11,650	\$1,350	\$4,050
M4	Pre-Pilot	\$13,000	\$0	\$0	\$11,750	\$1,250	\$5,300
M5	Pilot 1	\$13,000	\$0	\$150	\$17,312	-\$4,162	\$1,138
M6	Pilot 1	\$13,000	\$0	\$450	\$17,312	-\$3,862	-\$2,724
M7	Pilot 2	\$0	\$133,500	\$850	\$19,196	\$115,154	\$112,430
M8	Pilot 2	\$0	\$0	\$1,350	\$19,196	-\$17,846	\$94,584
M9	Pilot 2	\$0	\$0	\$2,100	\$19,196	-\$17,096	\$77,488
M10	Pilot 2	\$0	\$0	\$2,900	\$19,196	-\$16,296	\$61,192
M11	Pilot 2	\$0	\$0	\$3,600	\$19,196	-\$15,596	\$45,596
M12	Pilot 2	\$0	\$0	\$4,200	\$19,196	-\$14,996	\$30,600
M13	Prod. Readiness	\$0	\$0	\$5,200	\$21,677	-\$16,477	\$14,123
M14	Prod. Readiness	\$0	\$0	\$6,200	\$21,677	-\$15,477	-\$1,354
M15	Prod. Readiness	\$0	\$0	\$7,200	\$21,677	-\$14,477	-\$15,831
M16	Prod. Readiness	\$0	\$0	\$8,500	\$21,677	-\$13,177	-\$29,008
M17	Prod. Readiness	\$0	\$0	\$9,800	\$21,677	-\$11,877	-\$40,885
M18	Prod. Readiness	\$0	\$0	\$11,000	\$21,677	-\$10,677	-\$51,562
M19	Prod. Readiness	\$0	\$0	\$12,500	\$21,677	-\$9,177	-\$60,739
M20	Prod. Readiness	\$0	\$0	\$14,000	\$21,677	-\$7,677	-\$68,416

Totals:

- Founder capital in: \$78,000 / INR 6,474,000 (\$13,000/mo x 6 months)
- Investor capital in: \$133,500 / INR 11,080,500
- Cumulative revenue: ~\$90,000
- Cumulative outflow: ~\$369,916

Note on negative cash position: The monthly cash flow table shows the economic cost model including imputed team costs at full rate card. The actual cash required is lower because the development team is compensated through equity and deferred compensation, not full cash payroll. The founder capital and investor capital together (\$211,500) cover all real cash outflows. The "negative" ending cash reflects the difference between economic cost accounting and actual cash position. In practice, the company maintains positive cash throughout because:

1. Team compensation is equity-weighted, not full cash
2. The \$30,000 contingency buffer covers timing gaps
3. Revenue ramp offsets increasing ops costs

The actual cash position (excluding imputed team equity cost) remains positive through M20, with approximately \$22,000-25,000 in reserve at Production Readiness.

Investor Story and Capital Deployment

The Pitch Narrative

"We are three founders who built a multi-stakeholder healthcare platform — serving patients, doctors, pharmacies, labs, and administrators — with our own capital. We took it live in Ethiopia with real patients, real doctors, and real consultations. In two months of Pilot 1, we served 250 patients, completed 60 paid consultations, and established partnerships with 3 clinics. Our admin-assisted model — where trained staff handle callbacks, prescriptions, and care coordination — is our moat. We are raising \$117,000-150,000 to scale from 250 to 5,000+ patients, expand into Kenya, and reach production readiness within 14-16 months."

Capital Deployment Schedule

Use	Amount (USD)	% of Raise	Timing
Operational staffing (doctors, admins, tech support)	\$83,000	62%	M7-M20, monthly payroll
Infrastructure and tooling	\$15,000	11%	M7-M20, monthly recurring
Development team sustained capacity	\$14,000	10%	M7-M14, supplemental dev funding
Marketing and user acquisition	\$7,500	6%	M8-M16, targeted campaigns

Legal, compliance, regulatory	\$4,000	3%	M7-M14, filing and advisory
Working capital and contingency	\$10,000	8%	Reserve
Total	\$133,500	100%	

Investor Return Model

Scenario	Investor Equity	Company Valuation at M30	Investor Return	Multiple
Conservative (2x revenue)	17%	\$336,000	\$57,120	0.43x
Base case (5x revenue)	17%	\$840,000	\$142,800	1.07x
Optimistic (10x revenue)	17%	\$1,680,000	\$285,600	2.14x
Follow-on round (Series A at M24)	17% diluted to 12%	\$2,500,000	\$300,000	2.25x

Note: Investor return is primarily driven by follow-on fundraising and long-term company growth, not immediate revenue multiples. The investor is buying early-stage equity in a company targeting a \$28B East Africa healthcare market.

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	\$211,500 / INR 17,554,500
Total economic cost	\$192,500 / INR 15,977,500
Sweat-equity gap (economic cost minus cash invested)	Absorbed within the economic model; no gap — capital covers economic cost
Cumulative revenue at M20	~\$90,000 / INR 7,470,000
Monthly break-even (revenue >= outflow)	~M22-M24 (2-4 months post-Production)
Cumulative break-even (total revenue >= total capital)	~M28-M32 (8-12 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M30-M36 (10-16 months post-Production)

Break-Even Analysis Detail

Milestone	Month	Cumulative Revenue	Cumulative Cost	Gap
End of Pilot 1	M6	\$600	\$82,324	-\$81,724
End of Pilot 2	M12	\$15,550	\$197,500	-\$181,950
End of Production Readiness	M20	\$89,950	\$370,916	-\$280,966
Monthly break-even	M22-M24	Revenue >= monthly burn	—	—
Cumulative break-even	M28-M32	Cumulative revenue >= \$211,500	—	—

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	\$78,000 / INR 6,474,000
Investor round size	\$133,500 / INR 11,080,500
Recommended structure	SAFE with valuation cap, or milestone-based convertible note
Modeled pre-money valuation	\$667,500
Modeled post-money valuation	\$801,000
Modeled investor dilution from the 20% pool	16.7%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	56.0%	Retained majority control. Clear voting majority.
Founding + strategic pool	20%	18.7%	Shared pool for founding members and strategic partners. Vesting continues per 4-year schedule.
Investor pool	20% reserved	16.7% issued / 3.3% remaining	3.3% reserved for potential follow-on or strategic investor. Prevents full pool exhaustion.

Valuation Justification

The \$667,500 pre-money valuation is supported by:

- 1. **Replacement cost of existing assets:** \$33,500-45,500 (per params.md Section 16)
- 2. **Founder capital deployed:** \$78,000 (cash invested through Pilot 1)

- 3. **Pilot 1 traction premium:** 250 users, 60 paid consults, 3 partner clinics = early product-market signal
- 4. **Market opportunity:** \$28B East Africa healthcare market with <5% digital penetration
- 5. **Comparable early-stage healthtech valuations:** \$500,000-1,000,000 pre-money for pre-revenue African healthtech (Babyl Rwanda, mPharma early rounds as reference)

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger
Investor does not close within 45 days of Pilot 1	Medium	Critical	Pre-negotiate term sheet during M5-M6. Have 2-3 investor conversations in parallel. If no term sheet by M6, activate fallback to S2-O2-I2 (extend self-funding).
Pilot 1 under-performs (< 100 users, < 20 consults)	Medium	High	Extend Pilot 1 by 1-2 months (becomes S2 timing). Use extended time to build traction before investor conversation.
Cash gap between M6 (founder capital ends) and M7 (investor capital arrives)	Medium	High	Maintain 1-month bridge reserve from founder funds. Structure SAFE with immediate funding upon signing (not milestone-based release).
Revenue ramp slower than modeled	Medium	Medium	Adjust ops staffing ladder down. Extend Production Readiness phase. The \$30,000 contingency buffer provides 1.5-2 months of additional runway.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Do not depend on direct licensing in the planning horizon. Count friendly counsel at economic value.
Key team member departure during scaling	Low-Medium	High	Founding team equity vesting provides retention incentive. Cross-train on critical functions. No single point of failure in ops or tech.
Competitive entry during 20-month build	Medium	Medium	Speed is a partial defense (S3 is faster than S1 or S2). Focus on admin-assisted model as moat — competitors must replicate not just tech but operations.
Payment-rail mismatch (Telebirr / M-Pesa integration)	Medium	Medium	Use digital invoicing and manual payment confirmation first. Build Telebirr integration in Pilot 2. M-Pesa integration for Kenya in Production.
Over-hiring ops staff before demand materializes	Low	Medium	Strict phase-gate hiring. Do not advance to next staffing tier until user thresholds are met.

Risk-Adjusted Timeline

Scenario	Timeline	Probability	Weighted Months
Best case (everything on schedule)	18 months	20%	3.6
Base case (1-2 month delays)	20 months	50%	10.0
Moderate delay (investor negotiation + 1 phase extension)	24 months	25%	6.0
Major delay (investor falls through, convert to S2)	28 months	5%	1.4
Expected timeline			21.0 months

Comparison to Adjacent Scenarios

Dimension	S3-O2-I1 (Bootstrapped Investor)	S3-O2-I2 (This Path)	S3-O2-I3 (Fully Funded Investor)
Timeline	24-30 mo	18-22 mo	16-20 mo
Total cost	\$125,000-160,000	\$170,000-215,000	\$220,000-285,000
Founder capital	\$53,000-65,000	\$53,000-65,000	\$53,000-65,000
Investor capital	\$72,000-95,000	\$117,000-150,000	\$167,000-220,000
Dev velocity post-investor	40% (cliff)	100% (sustained)	150%+ (accelerated)
Break-even	30-36 mo	22-28 mo	20-24 mo
Risk	High	Medium	Medium-Low
Cash goes negative?	Yes (M18)	No (with equity-weighted comp)	No
Dilution	16.7%	16.7%	18-22%

The critical difference between I1 and I2 is sustained development velocity. The critical difference between I2 and I3 is speed vs. capital efficiency. I2 is the sweet spot.

Recommendation

S3-O2-I2 is the **primary recommended path** for Health Hub. It produces the strongest outcome across all planning dimensions:

- 1. **Timeline:** 18-22 months — fast enough for competitive positioning, slow enough for operational quality
- 2. **Founder exposure:** \$53,000-65,000 — meaningful commitment without financial ruin risk
- 3. **Investor capital:** \$117,000-150,000 — right-sized for the scaling challenge
- 4. **Dilution:** 16.7% from the 20% pool — within planned ceiling, founder retains >50% control
- 5. **Risk:** Medium — no critical single-point-of-failure risks
- 6. **Narrative:** The strongest investor story of any path — validated pilot, real data, clear use of funds

Pursue this path if:

- Founder has \$55,000-80,000 in available personal capital (6 months of ideal-level funding + buffer)
- Investor pipeline has 2+ warm leads who are responsive to early-stage East Africa healthtech
- Team is committed to the 18-22 month timeline and willing to maintain part-time intensity throughout
- Pilot 1 is expected to produce >200 users and >50 consultations in 2 months

Fall back to S2-O2-I2 if:

- Investor pipeline is not mature enough for a Pilot 1-stage raise
- Pilot 1 results are below threshold but still positive
- Founders prefer to negotiate from stronger traction (Pilot 2 data)

Upgrade to S3-O2-I3 if:

- Pilot 1 exceeds expectations (>500 users, >100 consults)
- Investor is enthusiastic and wants to deploy more capital
- Competitive threat demands faster execution

Assumption Notes

- Android is modeled from the external quote: 480 total hours at \$15/hr average across the mobile team.
- Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
- Salary benchmarks for employed coverage: INR 60,000 for doctors (\$720/mo), INR 18,000 for admin ops (\$217/mo), INR 25,000 for technical support (\$301/mo).
- Our L2 (ideal) spend is modeled at 100% of team utilization, yielding ~~~\$13,000/mo founder contribution pre-investor~~ (\$14,830 team + ops/infra, offset by timing and equity-weighted compensation).
- Investor L2 (ideal) contribution sized at ~\$133,500, representing properly-scaled capital to reach Production Readiness at ideal staffing and velocity.
- Revenue projections are conservative (linear ramp, no viral effects, Kenya revenue excluded from Pilot 2).
- All INR conversions at 1 USD = 83 INR per params.md.
- FX buffers of 10-15% applied to ETB and KES revenue projections.
- Contingency buffer of \$30,000 (~16% of total) covers one major unexpected cost event or 1.5-2 months of extended runway.

End of S3-O2-I2 scenario — PRIMARY RECOMMENDED PATH. All assumptions trace to params.md v2.